Infinite family of rank ≥ 7

Problèmes ouverts — A. Dujella · 2026

Problem 2.7 (Dujella, 2026). Is there an infinite family of elliptic curves, parametrized by rational points of a genus 1 curve, with torsion group $\mathbb{Z}/4\mathbb{Z}$ and rank $\geqslant 7$?

Remark. There is an elliptic curve over $\mathbb{Q}(t)$ with torsion group $\mathbb{Z}/4\mathbb{Z}$ and rank 6 (see Dujella, A., Peral, J.C. : An elliptic curve over $\mathbb{Q}(u)$ with torsion $\mathbb{Z}/4\mathbb{Z}$ and rank 6. Rad Hrvat. Akad. Znan. Umjet. Mat. Znan. **28**, 185–192 (2024)).